

2024 Calculus II Additional Exercise (5)

1. Exercise 2 in page 32 from Calculus Supplement Homework.
2. Let C be a parametric curve given by the equation $x = a(t - \sin t)$, $y = a(1 - \cos t)$, $t \in [0, 2\pi]$.
 - (a) Find the area enclosed by the curve C and the x -axis.
 - (b) Find the length of the curve C .
 - (c) Find the area of the surface generated by revolving the curve C about x -axis.
 - (d) Find the centroid of the curve C .
3. Let C be given by the equation $r^2 = a^2 \cos 2\theta$, $\theta \in [-\frac{\pi}{4}, \frac{\pi}{4}]$.
 - (a) Graph the curve C in the Cartesian xy -plane.
 - (b) Find the area enclosed by the curve C .
4. Exercise 4 in page 32 from Calculus Supplement Homework.
5. Exercise 12 in page 33 from Calculus Supplement Homework.

Q1, P32 Ex2

2. Find the open intervals of x such that the curve $y = f(x)$ is concave down, where the curve is given by:

$$\begin{cases} x = t^3 + 3t + 1 \\ y = t^3 - 3t + 1 \end{cases}.$$

Sol.

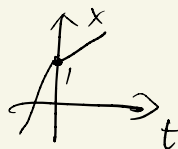
concave down: $\frac{dy}{dx}$ decrease on I .

$$\frac{dy}{dx^2} < 0$$

$$\begin{cases} x = t^3 + 3t + 1 \\ y = t^3 - 3t + 1 \end{cases} \Rightarrow \begin{cases} x'(t) = 3t^2 + 3 \\ y'(t) = 3t^2 - 3 \end{cases} \Rightarrow \frac{dy}{dx} = \frac{t^2 - 1}{t^2 + 1} = \frac{-2}{t^2 + 1}$$

$$\frac{d^2y}{dx^2} = \frac{dy'}{dt} \cdot \frac{dt}{dx} = \frac{4t}{(t^2 + 1)^2} \cdot \frac{1}{3t^2 + 3} = \frac{4t}{3(t^2 + 1)^2(t^2 + 1)} < 0$$

Sol. $t < 0 \Rightarrow x < 1$ by picture



□

Q2

2. Let C be a parametric curve given by the equation $x = a(t - \sin t)$, $y = a(1 - \cos t)$, $t \in [0, 2\pi]$.

(a) Find the area enclosed by the curve C and the x -axis.

(b) Find the length of the curve C .

(c) Find the area of the surface generated by revolving the curve C about x -axis.

(d) Find the centroid of the curve C .

Sol:

弧长微元

$$\begin{cases} x = a(t - \sin t) \\ y = a(1 - \cos t) \end{cases} \Rightarrow \begin{cases} \frac{dx}{dt} = a(1 - \cos t) \\ \frac{dy}{dt} = a \sin t \end{cases} \Rightarrow dl = \sqrt{[x'(t)]^2 + [y'(t)]^2} dt$$

$$= |a| \sqrt{2 - 2\cos t} dt$$

$$= 2|a| \sin \frac{t}{2} dt$$

$$(a) A = \int_0^{2\pi} y dx$$

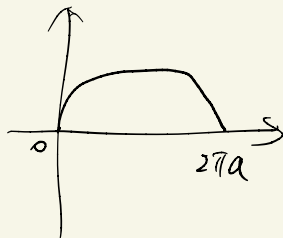
$$= \int_0^{2\pi} a^2 (1 - \cos t)^2 dt$$

$$= a^2 \int_0^{2\pi} 1 - 2\cos t + \cos^2 t dt$$

$$= a^2 \int_0^{2\pi} 1 - 2\cos t + \frac{1 + \cos 2t}{2} dt$$

$$= a^2 \left[\frac{3}{2} t - 2 \sin t + \frac{\sin 2t}{4} \right] \Big|_0^{2\pi}$$

$$= 3\pi a^2$$



(b)

$$L = \int_0^{2\pi} \sqrt{[x'(t)]^2 + [y'(t)]^2} dt$$

$$= \int_0^{2\pi} \sqrt{a^2(1 - \cos t)^2 + a^2 \sin^2 t} dt$$

$$= |a| \int_0^{2\pi} \sqrt{2 - 2\cos t} dt$$

$$= |a| \int_0^{2\pi} \sqrt{4 \left[\sin \frac{t}{2} \right]^2} dt$$

$$= 2|a| \int_0^{2\pi} \sin \frac{t}{2} dt$$

$$= -4|a| \cos \frac{t}{2} \Big|_0^{2\pi}$$

$$= 8|a|$$

$$(c) S = \int 2\pi |y| dl$$

$$= \int_0^{2\pi} 2\pi |y| \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

表面积微元
 $dS = 2\pi |y| dl$

$$= \int_0^{2\pi} 2\pi |y(t)| \sqrt{[x'(t)]^2 + [y'(t)]^2} dt$$

$$= 2\pi \int_0^{2\pi} |a|(1 - \cos t) \cdot 2|a| \sin \frac{t}{2} dt$$

$$= 4\pi a^2 \int_0^{2\pi} 2 \sin^3 \frac{t}{2} dt$$

$$= -16\pi a^2 \int_0^{2\pi} 1 - \cos^2 \frac{t}{2} d \cos \frac{t}{2}$$

$$= -16\pi a^2 \left(u - \frac{u^3}{3} \right) \Big|_1^{-1}$$

$$= -16\pi a^2 \left(-1 + \frac{1}{3} - \left(1 - \frac{1}{3} \right) \right)$$

$$= \frac{64}{3} \pi a^2$$

(d)

$$L_x = \int_C x dl$$

$$= \int_0^{2\pi} a(t - \sin t) 2|a| \sin \frac{t}{2} dt$$

$$= 2a|a| \left(-2 \int_0^{2\pi} t d \cos \frac{t}{2} - \int_0^{2\pi} 2 \sin \frac{2t}{2} \cos \frac{t}{2} dt \right)$$

$$= 2a|a| \left(-2t \cos \frac{t}{2} \Big|_0^{2\pi} + 2 \int_0^{2\pi} \cos \frac{t}{2} dt - 4 \int_0^{2\pi} \sin \frac{2t}{2} d \sin \frac{t}{2} \right)$$

$$= 2a|a| \left(4\pi + 0 - 4 \cdot \frac{1}{3} \sin \frac{2t}{2} \Big|_0^{2\pi} \right)$$

$$= 8\pi a|a|$$

$$\begin{aligned}
L_y &= \int_C y \, dl \\
&= \int_0^{2\pi} a(1 - \cos t) \cdot 2a \left| \sin \frac{t}{2} \right| dt \\
&= 2a|a| \left(\int_0^{2\pi} \sin \frac{t}{2} dt + 2 \int_0^{2\pi} 2 \cos \frac{t}{2} - 1 \, d \cos \frac{t}{2} \right) \\
&= 2a|a| \left[2 \cos \frac{t}{2} \Big|_0^{2\pi} + 2 \cdot \left(\frac{2}{3} \cos \frac{3t}{2} - \cos \frac{t}{2} \right) \Big|_0^{2\pi} \right] \\
&= 2a|a| \left[4 + 2 \left(-\frac{2}{3} + 1 - \left(\frac{2}{3} - 1 \right) \right) \right] \\
&= 2a|a| \left[4 + \frac{4}{3} \right] \\
&= \frac{32a|a|}{3}
\end{aligned}$$

$$\therefore \tilde{x} = \frac{L_x}{L} = \frac{8\pi a|a|}{8|a|} = \pi a$$

$$\tilde{y} = \frac{L_y}{L} = \frac{\frac{32a|a|}{3}}{8|a|} = \frac{4}{3}a$$

D

Q3.

3. Let C be given by the equation $r^2 = a^2 \cos 2\theta$, $\theta \in [-\frac{\pi}{4}, \frac{\pi}{4}]$.

(a) Graph the curve C in the Cartesian xy -plane.

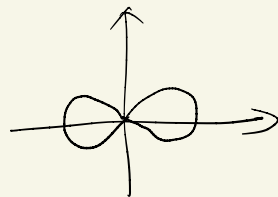
(b) Find the area enclosed by the curve C .

Sol:

$$r^2 = a^2 \cos 2\theta \quad \theta \in \left[-\frac{\pi}{4}, \frac{\pi}{4}\right] \quad (a)$$

$$r^4 = a^2 r^2 \cos^2 \theta - a^2 r^2 \sin^2 \theta$$

$$(x^2 + y^2)^2 = a^2 (x^2 - y^2)$$



(b)

极坐标面积微元 $dA = \frac{1}{2} r^2 d\theta$

$$A = 2 \int_C \frac{1}{2} r^2 d\theta$$

$$= a^2 \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \cos 2\theta d\theta$$

$$= \frac{a^2}{2} \sin 2\theta \Big|_{-\frac{\pi}{4}}^{\frac{\pi}{4}}$$

$$= a^2$$

Q4. P32 Ex4

4. Find the area of the surface generated by revolving the curve about the y-axis:

$$x = 2 \cos t, \quad y = 3 + \sin t, \quad 0 < t < 2\pi.$$

Sol.

$$\text{or } \begin{cases} x = 2 \cos t \\ y = 3 + \sin t \end{cases} \Rightarrow \begin{cases} x'(t) = -2 \sin t \\ y'(t) = \cos t \end{cases}$$

$$S = \int 2\pi x \, dl$$

$$= 2 \int_0^{\pi} 2\pi \cdot 2 \cos t \sqrt{4 \sin^2 t + \cos^2 t} \, dt$$

$$= 8\pi \int_0^{\pi} \sqrt{3 \sin^2 t + 1} \, d \sin t$$

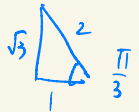
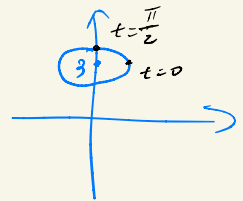
$$= 8\pi \int_0^1 \sqrt{3u^2 + 1} \, du$$

Let $u = \frac{1}{\sqrt{3}} \tan v$, $du = \frac{1}{\sqrt{3}} \sec^2 v$, where $\frac{\pi}{3} = \arctan \sqrt{3}$

$$= 8\pi \int_0^{\frac{\pi}{3}} \sqrt{\tan^2 v + 1} \cdot \frac{1}{\sqrt{3}} \sec^2 v \, dv$$

$$= \frac{8\pi}{\sqrt{3}} \int_0^{\frac{\pi}{3}} \sec^3 v \, dx$$

$$\left(\frac{x}{2}\right)^2 + (y-3)^2 = 1$$



Now compute $I = \int \sec^3 x dx = \int \sec x d \tan x = \sec x \tan x - \int \tan x d \sec x$
 $= \sec x \tan x - \int \sec x \tan^2 x dx = \sec x \tan x - \int \frac{1 - \cos^2 x}{\cos^3 x} dx$
 $= \int \sec x dx + \tan x \sec x - \int \sec^3 x dx$

$$\therefore I = \frac{1}{2} \left(\int \sec x dx + \tan x \sec x \right)$$

$$= \frac{1}{2} \left(\int \frac{\sec x (\sec x + \tan x)}{\sec x + \tan x} dx + \tan x \sec x \right)$$

$$= \frac{1}{2} \left(\ln |\sec x + \tan x| + \tan x \sec x \right) + C$$

$$\therefore \text{原式} = \frac{8\pi}{\sqrt{3}} \cdot \frac{1}{2} \left(\ln |\sec v + \tan v| + \tan v \sec v \right) \Big|_0^{\frac{\pi}{3}}$$

$$= \frac{4\pi}{\sqrt{3}} \left(\ln(2 + \sqrt{3}) + 2\sqrt{3} \right)$$

Q5 P33 Ex12

12

12. Assume that the curve is given by

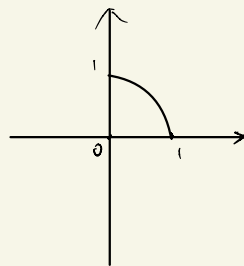
$$x = \sin t, \quad y = \cos(2t), \quad 0 \leq t \leq \frac{\pi}{2}.$$

Find the Cartesian equation for the curve and identify the graph. Further, find the area enclosed by the curve and x -axis and the volume of the solid generated by revolving the curve about x -axis.

Sol.

$$y = \cos(2t) = 1 - 2\sin^2 t = 1 - 2x^2$$

$$A = \int_0^1 y dx = \int_0^1 (1 - 2x^2) dx = \left[x - \frac{2x^3}{3} \right]_0^1 = \frac{1}{3}$$



$$\begin{aligned} V &= \int_0^1 dv = \int_0^1 \pi y^2 dx \\ &= \pi \int_0^1 1 - 4x^2 + 4x^4 dx \\ &= \pi \left(1 - \frac{4}{3} + \frac{4}{5} \right) \\ &= \frac{7}{15} \pi \end{aligned}$$

□